22 A cycle race across America is 3069.25 miles in length.

Juan knows his average speed for his previous races is 15.12 miles per hour.
For the next race across America he will cycle for 8 hours per day.

(a) Estimate how many days Juan will take to complete the race.

Round the distance to 3000 miles and speed to 15 mph.

Time cycling = 3000 ÷ 15 = 200 hours.

1 mark
Use rounded values in a correct calculation

	Miles	h	
×200	15	1	×200
	3000	200	

1 mark
Estimated time = 25 days

Days = 200 ÷ 8

$$8 \overline{)\,2^{2}0^{4}0} \quad = \quad 25$$

1 mark
Divide the cycling time by 8 hours per day

25

(3)

Juan trains for the race.
The average speed he can cycle at increases.
It is now 16.27 miles per hour.

(b) How does this affect your answer to part (a)?

Fewer days: a higher speed covers the same distance in less time.

1 mark
Explain the effect on time

(1)

(Total for Question 22 is 4 marks)

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